

MOHAMED AZAB

Ottawa, Ontario, Canada

📞 613-619-5360

✉️ mohazab2006@gmail.com

🌐 [LinkedIn](#)

🔄 [GitHub](#)

📁 [Portfolio](#)

EDUCATION

Carleton University School of Computer Science

Cumulative GPA: 10.80 / 12.00

Bachelor of Computer Science, AI & Machine Learning, Cyber Security | Minor in Business

Expected 2028

- **Relevant Coursework:** Data Structures, Algorithms, Systems Programming, OOP, Probability Models

EXPERIENCE

Junior DevOps Specialist

May 2026 – Aug 2026

CIRA (Canadian Internet Registration Authority) – Co-op

Ottawa, Ontario

- Supporting **cloud and on-prem infrastructure** for CIRA's cybersecurity and DNS services at national scale.
- Developing **CI/CD pipelines** and automation workflows using **GitLab CI/CD**, Terraform, Ansible, and AWS.
- Improving system reliability through **monitoring, alerting**, and runbooks across distributed systems.
- Maintaining **infrastructure-as-code** practices and configurations across cloud and on-prem environments.

Co-Founder & Developer

October 2023 – Present

KOVA

Ottawa, Ontario

- Built and deployed **production web applications** for multiple clients with scalable, high-performance architecture.
- Designed and integrated **AI-driven features** including chatbots, automation, and intelligent data pipelines.
- Developed full-stack systems with **authentication, RBAC dashboards**, and API integrations across core services.
- Managed **cloud infrastructure** using **Docker, NGINX, SSL/TLS**, domains, firewalls, and backups.
- Implemented **CI/CD pipelines** with **GitHub Actions** to automate testing, builds, and deployments.

PROJECTS

XGenius | Python, Next.js, FastAPI, Pandas, NumPy, scikit-learn, XGBoost, PostgreSQL, Docker

April 2026

- Built an **AI-powered World Cup prediction engine** using **machine learning, xG modeling**, and Monte Carlo simulation.
- Developed models using **Pandas** and **NumPy** to process data, predict **match outcomes**, and estimate **expected goals**.
- Engineered **large-scale tournament simulations** (10,000+ runs) to compute team progression and win probabilities.

KnowledgeMesh | Python, Next.js, FastAPI, PostgreSQL, pgvector, Redis, Docker, Ollama, LLM APIs

March 2026

- Architected **multi-service RAG platform** generating **citation-backed answers** grounded in source document passages.
- Built **Redis** worker for extract, chunk, embed with **pgvector**, **MMR reranking**, and **LLM routing** (OpenAI/Ollama).
- Developed **Next.js** dashboard for documents, **SSE streaming**, diagnostics, rate limiting, and **JWT-secured APIs**.

SentinelStack | Python, Next.js, FastAPI, scikit-learn, PostgreSQL, Docker

March 2026

- Built an **AI-driven SOC platform** detecting threats, fusing signals, analyzing **port exposure**, and automating response.
- Engineered feature extraction across **8 behavioral signals** per source, feeding **Isolation Forest** detection.
- Implemented **score fusion (LOW→CRITICAL)** with automated alerting, IP blocking, and **LLM-assisted triage**.

PathToOffer | TypeScript, Python, Next.js, FastAPI, SQLite, ReportLab

January 2026

- Implemented **NLP pipelines** to score resumes against job postings and surface targeted **ATS skill gaps**.
- Built **LLM prompt pipelines** to generate tailored resumes, cover letters, and interview questions per posting.
- Designed **REST API** with 10+ endpoints supporting scoring, learning roadmap generation, and **ATS-ready PDF export**.

Cadence (Desktop) | TypeScript, Rust, React, Tauri, SQLite, Tesseract.js

September 2025

- Built a **cross-platform desktop app** using **Tauri** with a **Rust** backend and **local-first SQLite** storage.
- Implemented **OCR-based syllabus parsing** with **LLM-assisted extraction** of deadlines, weights, and tasks.
- Developed **AI-driven planning features** to prioritize work and surface gaps based on academic performance.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, Bash, Rust

Developer Tools: Git, GitHub Actions, GitLab CI/CD, Docker, Linux, Terraform, Ansible, AWS, NGINX, UFW, Tailscale

Frameworks & Libraries: Next.js, React, Node.js, FastAPI, Spring Boot, Tailwind CSS, XGBoost, JDBC, Axios, Vite

Databases & Storage: PostgreSQL, MySQL, SQLite, Supabase, pgvector, Redis

AI & ML: LLM APIs, Ollama, scikit-learn, XGBoost, Pandas, NumPy